

Summary

I am a fourth-year undergraduate student in Engineering Mechanics at Zhejiang University, working under the supervision of Shiyong Xiong. My research interests lie in scientific machine learning, reduced-order physical simulation, neural operators for PDEs, and computational fluid dynamics. I have also dabbled in quantum simulation and atomic physics.

Education

- Zhejiang University, Engineering Mechanics** Sep 2023 - Present
GPA: 4.07/4.3 | Ranking: 2/52

Paper

- Local Gradient Neural Operator** *Preprint*
Baiming Zhang, Jinsong Tang, Ying Xu*, Lihua Chen, Shiyong Xiong*
- Learnable Composition for Neural Operators** *Preprint*
*Zituo Chen, Baiming Zhang, Sili Deng**
- Gauge-Canonical Eigenfluids: Geometry-Conditioned Neural Spectra** *Preprint*
*Baiming Zhang, Zhecheng Wang, Shiyong Xiong, Eitan Grinspun**

Experience

- Fluid Mechanics & Machine Learning** Jan 2025 – Present
Under Mentor Shiyong Xiong
- Quantum States & Devices Research Group** Oct 2023 – Dec 2024
Under Prof. Dawei Wang & Dr. Xingqi Xu

Selected Awards and Honors

- International College Student Engineering Mechanics Competition, World Gold Medal**
First student in ZJU-SAA history to win the gold; featured on the school homepage.
- Chinese National Zhou Peiyuan Mechanics Competition, First Prize
- National Mechanics “101 Plan” Undergraduate Academic Forum, Outstanding Oral Presentation

Other Experience

- Teaching Assistant, Chinese Physics Olympiad Training Camp, 2024
- Awarded “Five-Star Volunteer” — the university’s highest honor for public service.
- Served as volunteer coordinator for the 2024 International Symposium on Quantum Computing and Quantum Optics.

Technologies

- Languages:** Python, MATLAB, HTML, LaTeX, etc.
- Platform:** PyTorch, Github, Houdini, Solidworks, Raspberry Pi, etc.
- Machine Tools:** Lathe, Milling machine, Shaper, Slotter, etc.